

Perovskite Photovoltaics in August 2026: Non-radiative recombination control and buried-interface treatment

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Abstract

Across a monthly literature corpus of 531 publications, 144 were examined closely to evaluate defect passivation, contact chemistry, phase stability, and operational durability in perovskite solar cells (PSCs). Independent certification establishes the performance frontier for monolithic tandem architectures at a power conversion efficiency (PCE) of 33.66%, driven by complementary bandgap absorption that suppresses thermalization losses alongside optimized current matching. Inverted single-junction configurations established a certified PCE ceiling of 27.4%, achieved primarily through tailored self-assembled monolayers (SAMs) and molecular additives that passivate undercoordinated lead defects and suppress interfacial molecular clustering. Scaling these passivated contacts to expanded dimensions incurs an area penalty governed by transparent conducting oxide sheet resistance, interconnect scribe dead area, and spatial film non-uniformity, yielding an accredited PCE of 17.35% at the largest reported aperture area of 61.2 cm². Operational longevity evidence remains sparse and disconnected from efficiency milestones. Only 5 investigations quantified operational lifetime, with a maximum duration of 1400 h, while only 2 studies adhered to standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols. Degradation pathways identified across this cohort include interfacial redox reactions between acidic phosphonic acid anchoring groups and polar processing solvents, photo-induced halide oxidation and ion migration, and residual lattice microstrain that accelerates phase segregation under continuous operation. Reporting practices reveal a pronounced discrepancy between self-reported claims and certified performance, as 55.5% of abstracts stated an efficiency value, whereas only 9.7% stated independent certification and only 18 articles reported verified values. Intermediate-phase modulation and seed-layer crystallization mitigate interfacial voids and non-radiative recombination at hole-selective contacts during initial deposition. Published evidence fails to confirm that these strategies arrest field-driven halide redistribution or contact degradation during prolonged operation. Resolving these bottlenecks requires operando chemical characterization under bias combined with standardized ISOS testing on scalable apertures.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite;

tandem solar cells; operational stability

1. Introduction

Perovskite photovoltaic research has moved beyond maximizing unencapsulated power conversion efficiency (PCE) on sub-millimetre laboratory apertures. The strongest mechanistic consensus demonstrates that coordinating Lewis-base ligands and self-assembled monolayers (SAMs) passivate non-radiative recombination centres at buried contacts by coordinating undercoordinated lead sites. Yet the critical open problem has shifted to operational lifetime, device area, and rigorous verification. Arresting field-driven halide migration, phase segregation, and contact corrosion during prolonged continuous operation across scalable apertures remains unresolved, presenting barriers that transient efficiency improvements cannot circumvent.

To evaluate these challenges, 531 indexed works met the scope gate in August 2026 and 144 were read closely.

The resulting evidence base is structured across core physical domains: defect tolerance, passivation and ion migration, buried interfaces and contact chemistry, composition, phase control and crystallisation, wide-bandgap absorbers and tandem integration, as well as operational stability and degradation pathways.

Throughout each topic, this review distinguishes what the month’s literature verified from what it asserted. Theoretical passivation mechanisms and charge-transport claims are scrutinized directly against empirical device behavior, separating documented operational survivability from unverified assertions based on uncertified, small-area cells subjected to brief, non-standardized aging routines.

2. The Certified Frontier in August 2026

Independent certification establishes that spectral splitting across complementary bandgaps yields the highest verified efficiencies. Suppressed thermalization and optimized current matching drive this outcome in two-terminal (2T) perovskite tandems. Verified 2T power conversion efficiency (PCE) benchmarks reached a ceiling of 33.66%, followed by certified reports at 32.56%, 30.77%, 29.7%, and 27.84% [1,2,3,4,5]. Single-junction inverted (p-i-n) devices, constrained by non-radiative interfacial recombination and extraction barriers at selective contacts, established a narrower certified ceiling between 26.77% and 27.4%, led by verified values of 27.4%, 27.2%, and 26.77% [6,7,8]. Despite these verified milestones, rigorous certification remains rare across the literature: only 18 of 144 closely read papers reported an independently certified efficiency, allowing uncertified self-reported champion values to run ahead of verifiable baselines.

Translating these contact interfaces across larger dimensions exposes severe transport bottlenecks governed by transparent electrode sheet resistance, scribe line geometric dead area, and spatial film non-uniformity. At expanded dimensions, minimodule architectures maintained reported efficiencies of 27.7% on 25 cm² and 27.01% on 58.51 cm² [9,10]. However, larger apertures revealed significant performance losses, dropping to 23.3% on 57.3 cm², 23.14% on 29.7 cm², and 17.35% on 61.2 cm² [11,12,13]. A champion value recorded on a sub-0.1 cm² laboratory cell isolates microscopic interfacial transport under near-ideal current extraction, whereas an independently certified value measured across a module aperture gauges lateral conduction, spatial defect distributions, and coating reproducibility. These two metrics represent entirely distinct categories of physical evidence, meaning small-cell records cannot demonstrate module viability.

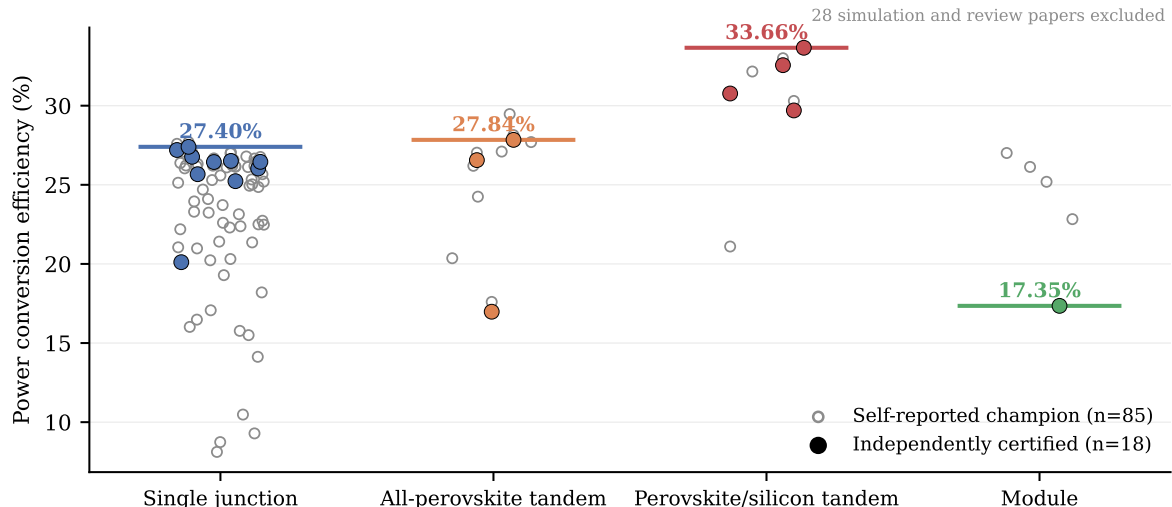


Figure 1. The certified frontier in August 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

3. Defect Tolerance, Passivation and Ion Migration

Defect formation and ion migration in metal halide perovskites are coupled through mobile halide vacancies that trigger operational degradation. Continuous illumination accelerates iodide oxidation to molecular iodine, generating iodine vacancies that expose undercoordinated lead (Pb^{2+}) sites [14], whereas nonadiabatic molecular dynamics shows that iodine vacancies drive hydrogen vacancy formation into stable VI · VH defect complexes whose migration is immobilized by rigid Pb-C coordinate bonding in VI · VH-C configurations [15]. Kinetic and compositional regulation confirms that mobile halide management is prerequisite to suppressing ion drift [16,17]. Exploiting the kinetic accessibility of bromide (Br^-) diffusion, an ordered anion-cation modification (OACM) strategy repairs anionic vacancies before cation passivation, suppressing phase segregation to achieve a 1.33 V open-circuit voltage (V_{oc}) and 19.21% stabilized efficiency in 1.83 eV wide-bandgap (WBG) top cells (champion 26.2% tandem efficiency) [16]. Selenocyanate pseudohalide alloying similarly retards ion drift and homogenizes formamidinium (FA^+) distributions throughout 1.66 eV absorbers [17]. Structural barriers provide complementary confinement. Scaffold-directed polyfluorinated 7F-EA-HI forms an interwoven tilted one-dimensional/three-dimensional (1D/3D) heterointerface that blocks defect propagation and ion migration in inverted (p-i-n) devices reaching an uncertified 24.94% power conversion efficiency (PCE) [18]. Self-driven halide exchange with bromide-rich nanocrystals passivates iodide vacancies via a surface-graded heterojunction, though restricted to an uncertified 22.19% champion PCE in an unreviewed preprint [19].

Beyond mobile halides, eliminating deep non-radiative traps demands coordinated passivation of undercoordinated metal cations and organic cation vacancies [14,20,21,22]. Bulk phenylthiourea coordinates Pb^{2+} to retard ion migration and suppress defect formation under illumination [14]. Ammonium alginate suppresses FA^+ deprotonation while passivating undercoordinated Pb^{2+} , raising fill factor (FF) from 80.2% to 85.1% in an uncertified 26.12% champion cell [20]. Dual-site coordination similarly targets opposing charges: sodium cyclamate coordinates Pb^{2+} via sulfonic acid groups and binds iodide vacancies through amino hydrogen bonds [21], while multidentate sulfamethoxa-

zole coordinates Pb^{2+} , organic-cation, and halide vacancies, lifting normal (n-i-p) efficiency from 18.01% to an uncertified 21.05% champion (20.31% average) [22]. Coupled hydrogen-bonding and electrostatic networks from the ionic liquid BSPH coordinate FA^+ , halides, and Pb^{2+} , suppressing trap density and ion drift to yield a 23.72% p-i-n PCE [23]. In hole-conductor-free carbon-electrode mesoscopic cells, DMSC passivates both Pb and I species to reach a 22.5% champion PCE [24], while multi-pyridyl spirobifluorenes coordinate defects during crystallization to yield 20.98% PCE versus 18.67% in controls [25]. In tin-based absorbers prone to Sn^{2+} oxidation, triphenylphosphine oxide coordinates Sn^{2+} as a hard Lewis base, yielding 14.13% PCE and a T80 of 720 h under unlabelled testing conditions [26].

A third frontier links defect formation directly to lattice mechanics and buried transport contacts [27,28,29]. Solid-solution lattice hardening by partial lanthanum substitution of Pb raises the Young modulus from 27.96 to 35.71 GPa, demonstrating that rigidifying the inorganic cage suppresses defect formation to yield a 26.54% champion PCE and a certified steady-state efficiency of 26.03% [27]. Dynamic strain homogenization via in situ supramolecular networks modulates Sn^{2+} coordination in mixed tin-lead absorbers, suppressing segregation to retain 84.7% initial efficiency after 450 h maximum power point tracking (MPPT) under one sun and 92.1% after 50 thermal cycles (24.1% champion PCE) [28]. At buried contacts, Ace-K bidentately chelates uncoordinated Sn^{4+} and oxygen vacancies on tin dioxide (SnO_2) while outcompeting dimethyl sulfoxide (DMSO) for lead iodide (PbI_2), eliminating voids to retain 92.4% efficiency after 1000 h MPPT on a 0.0729 cm^2 cell (26.17% champion PCE) [29]. Interfacial molecular bridging with 2-formylbenzenesulfonic acid sodium salt connects self-assembled monolayers (SAM) to perovskite, delivering an uncertified 26.21% champion PCE with an 86.15% FF [30]. At module scale, potassium dihydrogen phosphate disaggregates SnO_2 colloids and passivates buried contacts, yielding 23.16% PCE on 12 cm^2 modules with 8% scaling loss from a 25.34% champion cell [31].

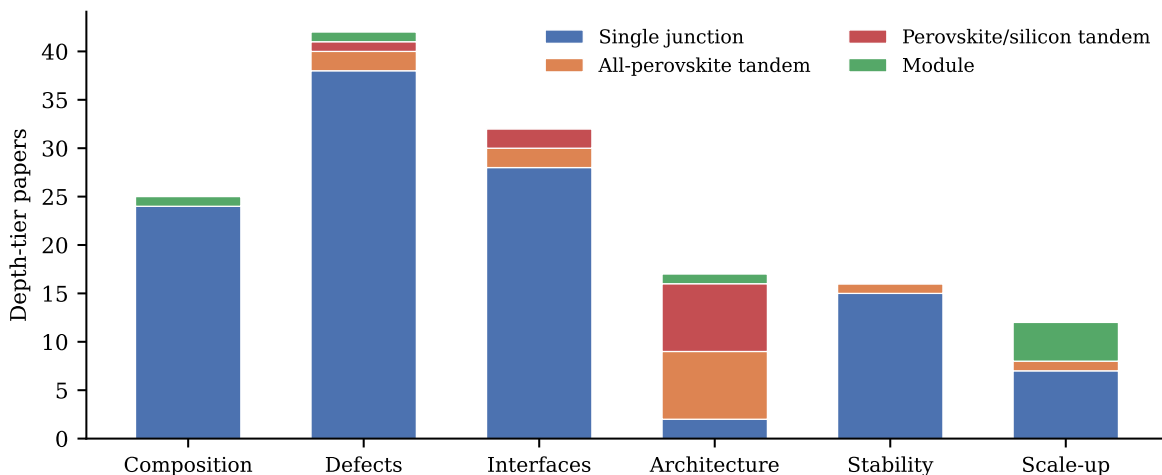


Figure 2. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

4. Buried Interfaces and Contact Chemistry

At buried hole-selective contacts in inverted positive-intrinsic-negative (p-i-n) and tandem perovskite solar cells (PSCs), disordered molecular packing and aggregation in self-assembled monolayers

(SAMs) cause severe pinhole shunts and energy misalignment. Suppressing self-assembled monolayer (SAM) clustering while keeping anchoring sites accessible resolves these losses: non-anchoring additives like TTMB secure open binding sites to reach a certified power conversion efficiency (PCE) of 33.66% (33.66% stabilised) in two-terminal (2T) perovskite/silicon tandems [1]. Hydrogen-bond networks via glycine hydrochloride in [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-4PACz) yield 22.38% champion PCE in pure-iodide wide-bandgap cells [32]. DMPA-based multilayers achieve a certified 27.2% PCE (27.59% champion) with 94.5% retention after 1600 hours under continuous 1 Sun illumination at 65 °C [7]. Interfacial dipole engineering further optimizes band offsets. Depositing bismuth oxychloride (BiOCl) onto SAMs induces dipole flipping and chlorine-iodine exchange, yielding a certified 27.84% 2T tandem PCE (28.14% champion, 27.86% stabilised), although single-junction efficiencies of 27.25% (1.55 eV) and 20.24% (1.85 eV) remain uncertified on small 0.09 cm² aperture areas [5]. This aligns with co-assembling 4-ASA with Me-4PACz on nickel oxide (NiO_x) to yield a 20.36% four-terminal (4T) tandem PCE (17.82% scanned single junction), and dipole tuning via m-FBC raising PCE from 23.05% to an uncertified 25.03%, though stability was tested only under 500 hours of uncharacterized storage [33,34]. Asymmetric heteroatom conjugation also regulates contact energetics: S-AS on NiO_x achieves 25.70% steady-state output (26.49% champion) and LW-1 on fluorine-doped tin oxide (FTO) reaches 26.0% stabilised efficiency (26.38% champion) [35,36].

Beyond electrostatic alignment, buried contact molecules template crystallization, relieve lattice strain, and passivate undercoordinated lead (Pb²⁺) defects. Multi-arm donor-acceptor dipoles (T4IA) coordinate Pb²⁺ through C=N and methoxy groups to relieve residual strain (26.79% champion PCE), whereas non-covalent templating between precursor additives (TPANI) and structurally matched triphenylamine bisphosphonic acid SAMs on indium tin oxide (ITO) reduces intergranular groove depth and reinforces adhesion (26.58% champion) [37,38]. Interfacial Pb-S bonding provides a parallel passivation route. Thienyl-based SAM additives template preferential (111) crystallization and segregate at grain boundaries to secure a certified 32.56% (33.0% champion) 2T tandem PCE, mirroring solution-phase sulfidation with dibutyl sulfide that induces an interfacial p-to-n transition to yield an uncertified 26.28% PCE [2,39].

Contact degradation also stems from chemical reactivity across metal oxide substrates. On NiO_x hole transport layers (HTLs), removing deleterious surface hydroxyls with ammonium thiocyanate (NH₄SCN) lowers transfer barriers and lattice strain (21.41% champion) [40]. Decoupling SAM anchoring from charge extraction through composite NiO_x stacks (sputtered base plus solution-deposited hydroxyl overlayer) permits scalable processing, achieving 24.86% PCE across 12.4 cm² [41]. Enamine SAM anchor engineering is consistent with this, achieving 23.14% PCE with 83.11% fill factor (FF) on a 29.7 cm² mini-module (>25.5% on small cells) [12]. At tin oxide (SnO₂) electron transport layers (ETLs) in regular negative-intrinsic-positive (n-i-p) devices, native surface hydroxyls and oxygen vacancies provoke interfacial void formation and microcracking. Converting hydroxyls into covalently anchored acetate via ethanol vapor reconstruction suppresses both (26.19% champion) [42]. Subliming pyruvic acid at 145 °C purges defect sites instead, reaching 26.17% PCE on 1 cm² with an 80% operational lifetime (T80) of 900 hours [43]. However, interfacial engineering reveals a trade-off between extraction and mechanical integrity: although fullerene SAMs (C60-PA) on SnO₂ raise initial tracked efficiencies from 18.6% to a 20.0% stabilised value, performance degrades under illumination at elevated temperatures because fullerenes weaken adhesion at the oxide interface [44].

5. Composition, Phase Control and Crystallisation

Inhomogeneous nucleation triggers point defects, phase segregation, and structural disorder in metal halide perovskite absorbers, requiring precise crystallization modulation [45]. Controlling intermediate solvent-solute coordination dictates lattice orientation and residual stress. Destabilizing solvent-PbI₂ adducts through cooperative interactions between sodium hydroxyquinoline sulfonate and sodium aminonaphthalene sulfonate drives out-of-plane [100] growth while converting tensile strain into compressive strain. This yields an uncertified champion power conversion efficiency (PCE) of 26.21% [46]. Similarly, dual-mode coordination with 4-chlorophenethylammonium (4Cl-PA) generates a macroporous PbI₂ template that facilitates organic cation infiltration. Carrier lifetime extends to 161 ns by time-resolved photoluminescence (TRPL) in uncertified 22.73% efficient devices [47]. In tin perovskites, intermediate-mediated stepwise crystallization during a pre-annealing static stage with fluorinated phenethylammonium alters the [SnI₆]⁴⁻ electronic landscape to suppress Sn²⁺ oxidation, reaching a 9.29% PCE [48]. However, chemical coordination efficacy depends on the processing solvent. SnF₂ suppresses Sn²⁺ oxidation in dimethylformamide:dimethyl sulfoxide (DMF:DMSO). It offers limited antioxidant protection in DMSO-free DMF:2-methylaminoethanol (DMF:2MAP), where formamidinium tin iodide (FASnI₃) films remain inherently Sn²⁺-rich [49].

Phase stability is equally bounded by thermodynamic competition and local defect equilibria [50,51,52]. In cesium-based systems, excess PbI₂ suppresses Ruddlesden-Popper antiphase boundaries (RP-APBs) but triggers edge-sharing [PbI₆]⁴⁻ motifs that nucleate the non-photoactive delta phase. Dimethylammonium (DMA⁺) phase engineering circumvents this breakdown by forming beta-(DMA,Cs)PbI₃ and intrinsically eliminating RP-APBs, achieving an uncertified 20.23% PCE [50]. Similarly, incorporating 0.8 mol% fluorinated methylammonium iodide (FMAI) stabilizes the perovskite lattice against thermal degradation in a 25.67% negative-intrinsic-positive (n-i-p) device. The reported 2000 h operational stability lacks detailed testing protocols [51]. In low-dimensional heterostructures, organic cation evaporation during annealing drives a dimensional shift from low- to high-n quantum wells. Precursor octylammonium bromide (OABr) concentration predetermines the initial n = 1 phase distribution in 19.29% wide-bandgap cells [52]. Replacing insulating organic spacers with semiconducting diketopyrrolopyrrole cations enables bidirectional charge transfer with inorganic [PbI₄]²⁻ slabs. Electron mobilities reach up to $1.3 \times 10^{-3} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, although functional solar cell metrics were omitted [53].

In lead-free multinary lattices, accessing target crystal phases requires overcoming severe kinetic and thermodynamic barriers [54,55,56]. Sequentially evaporating methylbismuth iodide (MBI) bilayers at substrate temperatures as low as 40 °C requires post-deposition annealing above 200 °C to yield a monoclinic C2/c phase with (111) texture. Champion PCE nonetheless reaches only 0.53% [54]. In double perovskites, solid-state mechanosynthesis reveals strict ionic size selectivity. Liquid assistance drives Rb⁺ into the A-site and Na⁺ into the B-site of Cs₂AgInCl₆ [55]. Conversely, solvent-free hydro-solution precipitation directly produces cubic Fm-3m Cs₂AgBiBr₆ without high-temperature annealing [56]. This indicates that solution-phase ionic transport bypasses the kinetic barriers inherent to vapor-phase and solid-state transformations.

6. Wide-Bandgap Absorbers and Tandem Integration

Non-conformal coverage and dewetting of hole-selective self-assembled monolayers (SAMs) on textured silicon substrates generate interfacial voids and unreacted lead iodide (PbI₂), driving non-radiative recombination in two-terminal (2T) perovskite/silicon tandems [4,57,58]. Resolving this

buried morphological mismatch requires tailored nucleation and crystallization. A CsCl seed layer facilitates organic precursor incorporation across MeO-2PACz to suppress interfacial PbI_2 , yielding a certified 29.7% power conversion efficiency (PCE) [4]. Solvent-free sequential evaporation capitalizes on accelerated lead halide conversion on textured silicon to deliver a 27.0% stabilized PCE [59]. Post-deposition methylammonium thiocyanate (MASCN) healing similarly repairs valley-confinement defects, driving secondary crystal growth into full-thickness columnar grains. That device achieved a certified 30.77% PCE on a 1.164 cm^2 aperture and retained initial performance after 3400 hours of continuous ambient maximum power point tracking (MPPT) [3]. In the wide-bandgap (WBG) top absorber, phase segregation and halide migration govern bulk degradation. Elevating the ion migration energy barrier improves local structural stability, supporting an uncertified champion PCE of 33.02% [60]. Two-dimensional (2D) (APD) PbI_4 seed templates impart regular compressive strain into the three-dimensional (3D) lattice, suppressing phase separation for a 29.47% PCE [61]. Contact durability under subsequent processing presents a further constraint. Sacrificial HTFNA coordination shields SAMs against atomic layer deposition (ALD) damage, yet this certified 26.56% efficiency is restricted to a 0.062 cm^2 cell [62]. The field is shifting toward resilient inorganic contacts like nickel oxide with N719 dye, which preserve 96.4% PCE after one year of outdoor MPPT [63]. Translating these interfaces to large-area modules escalates multi-scale defect densities [64], requiring kinetic crystallization control. Polyfluoroarene additives elevate effective nucleation barriers to yield a 27.01% module PCE across 58.51 cm^2 with 55 mV non-radiative voltage loss [10]. Static antisolvent engineering matches this, achieving 27.7% on 25 cm^2 [9].

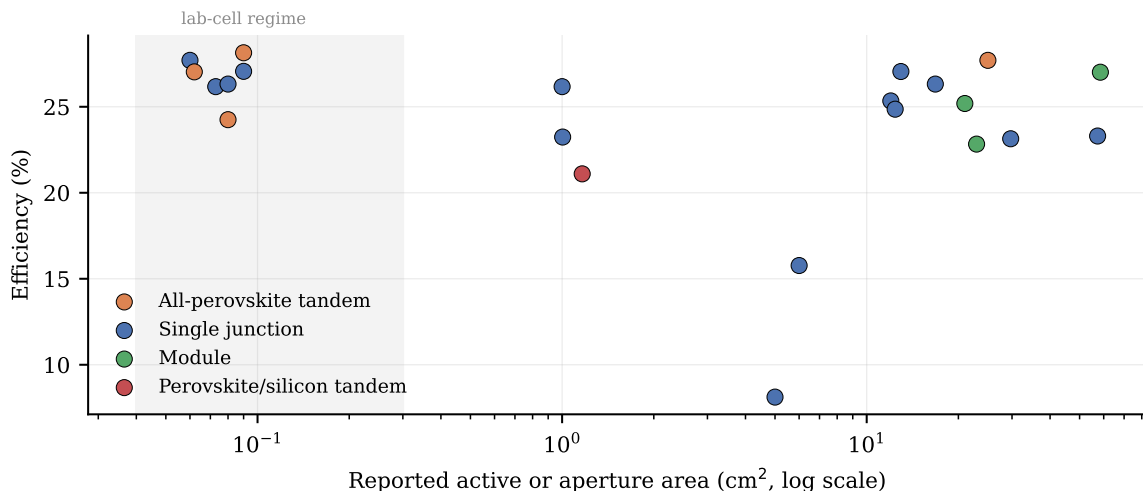


Figure 3. Efficiency against reported active or aperture area. The shaded band marks the sub- 0.3 cm^2 laboratory-cell regime in which most champion values are measured.

7. Operational Stability and Degradation Pathways

Interfacial chemical and photochemical degradation at selective contacts constitutes the primary failure mechanism addressed this month. Inverted perovskite solar cells (PSCs) suffer from coupled redox degradation. Acidity in carbazole-based phosphonic acid self-assembled monolayers (SAMs) activates dimethyl sulfoxide (DMSO)-mediated iodide redox reactions during ink deposition [6]. Photo-oxidation of formamidinium iodide (FAI) yields iodine radicals that catalyze phenyl-C61-butyric acid methyl ester (PCBM) dimerization alongside FAI deprotonation [65]. Cross-linking self-assembled

monolayer (SAM) headers in situ [66], co-assembling binary monolayers [67], or balancing dipole configurations [68,69] suppresses interfacial defect generation. Yet performance claims frequently exhibit severe certification discrepancies or rely on tiny device areas, such as a 27.06% uncertified champion dropping to 20.11% certified on 0.09 cm^2 [69] and a certified 27.4% power conversion efficiency (PCE) tested on 0.06 cm^2 [6].

In the bulk absorber, unrelaxed microstrain and defect formation accelerate phase degradation. Three routes relieve lattice strain: 2 mol% sodium (Na^+) interstitials [70], two-dimensional (2D) intermediate phase templating [71], and vertical lattice misalignment suppression [72]. Each increases defect formation energies for lead and iodide vacancies and prevents nonphotoactive phase transitions [70,71,72]. At rear contacts, a 5 nm chromium (Cr) barrier beneath non-gold electrodes curtails atomic interdiffusion [73]. Coordination modifiers prevent direct perovskite contact with metal oxides [74].

Standardized operational metrics remain conspicuously scarce. Across 144 examined papers, only 5 reported a time to 80% of initial efficiency (T80). Merely 2 adhered to International Summit on Organic Photovoltaic Stability (ISOS) protocols, specifically ISOS-L-1 [67] and ISOS-L-2 [65]. Consequently, isolated metrics such as an 800 h T80 [70] or an 864 h T80 on 0.08 cm^2 [74] reflect unstandardized evaluations rather than verified operational longevity.

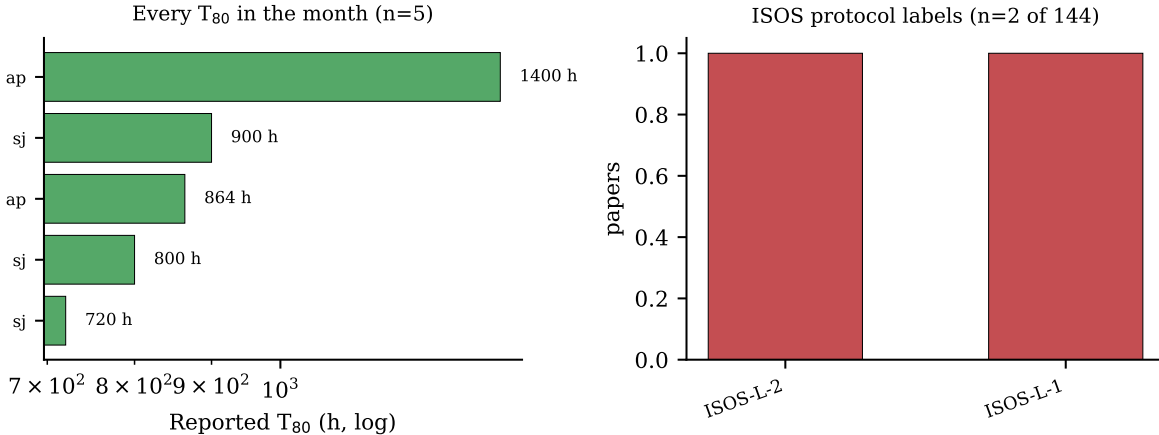


Figure 4. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

8. Research Gaps and Outlook

Controlling wide-bandgap (WBG) perovskite crystallization across textured silicon substrates represents the most supported mechanism this month, advancing certified two-terminal (2T) power conversion efficiency (PCE) to 33.66% [1] and 30.77% [3]. However, the assembled evidence leaves critical physical questions unestablished across four distinct areas.

First, although seed layers and secondary grain growth suppress phase segregation during deposition [3,4], current data do not demonstrate that columnar grains prevent electric-field-driven halide redistribution under sustained bias. Operando depth-profiling time-of-flight secondary ion mass spectrometry (ToF-SIMS) under operational voltage bias is required to determine whether seeded interfaces permanently impede ion migration or merely delay initial demixing. Second, non-anchoring

additives reduce aggregation in self-assembled monolayers (SAMs) [1], yet the literature fails to verify whether these modified contacts survive wet chemical processing. Acidic phosphonic acid SAMs trigger dimethyl sulfoxide (DMSO)-mediated iodide redox reactions during ink coating [6], leaving uncertified champion efficiencies vulnerable to severe certification discrepancies [69]. In situ grazing-incidence wide-angle X-ray scattering (GIWAXS) coupled with interface-sensitive photoelectron spectroscopy during solution wetting must establish whether molecular additives protect buried contacts from chemical degradation. Third, device scalability remains disconnected from small-area interfacial gains. Scale-up was addressed by 27 of 531 works, with the largest reported aperture reaching 61.2 cm² [13]. Reported data still do not decouple the physical origins of area-dependent efficiency penalties [71]. Calibrated absolute electroluminescence imaging resolving spatial ideality factors across square-decimeter modules is necessary to separate interconnect sheet resistance from spatial non-radiative recombination. Fourth, operational longevity remains unverified across the field. Only 5 papers reported a time to 80% of initial performance (T80) lifetime, with the longest reaching 1400 h [75]. Only 2 papers named an International Summit on Organic Photovoltaic Stability (ISOS) protocol [65,67]. Standardized ISOS testing combining elevated thermal stress with continuous maximum power point tracking (MPPT) under full illumination is needed to isolate contact delamination from intrinsic lattice decay.

The certified reporting share was 8.3 in 2026-07 and 9.7 in 2026-08, yet only 18 of 144 closely read papers reported any certified value. Requiring standardized ISOS protocols alongside certified aperture-area efficiencies would make next month’s literature directly comparable across laboratories.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

AI Usage Declaration

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extraction used qwen3.8-flash through the opencode interface; the prose was written by gemini-3.8-flash-high through the agy interface; independent review used Opus 5 at high reasoning effort. No figure is a model-generated image: every panel is plotted directly from the extracted records. Every quantitative statement in the main text resolves to the machine-generated statistics file shipped as Supplementary Information, and every citation resolves to a record carrying its source quotation. The author reviewed and edited the content, verified the selection, and takes full responsibility for the content of the published article.

References

1. Zeng, L. et al. Non-competitive additives regulate molecular assembly for compact monolayers in perovskite/silicon tandems. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-76266-0>
2. Zhang, D. et al. Thienyl Self-Assembled Molecule Enabled Stable Perovskite/TOPCon Silicon Tandem. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.5241611>
3. Kong, W. et al. Healing intervention for improving the efficiency and stability of tandem devices on industrial textured silicon. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-76713-y>
4. Škorjanc, V. et al. CsCl seed layer homogenizes co-evaporated perovskite growth for high-efficiency, fully textured perovskite-silicon tandem solar cells. *Joule* (2026). <https://doi.org/10.1016/j.joule.2026.102645>
5. Xie, Z. et al. Stacking-order-driven interfacial dipole flip and chlorine-iodine exchange for high-performance perovskite-organic tandem solar cells. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-76786-9>
6. Liang, Z. et al. Redirecting wet–interfacial redox pathways for efficient inverted perovskite solar cells. *Science* (2026). <https://doi.org/10.1126/science.aeg8415>
7. Pan, Y. et al. Self-Assembled Multilayers Reduce Interfacial Energy Loss in Perovskite Solar Cells. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-76497-1>
8. Feng, Y. et al. Ga Doping Enables (110) Facet Control and Surface Passivation of SnO₂ for Efficient and Stable Inverted Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.3095986>
9. Pallotta, R. et al. Static antisolvent processing for scalable and sustainable perovskite photovoltaics. *Communications Materials* (2026). <https://doi.org/10.1038/s43246-026-01321-8>
10. Zhang, T. et al. Synergistic Crystallization Regulation by Substituent Electronic Effects of Polyfluoroarenes Toward Large Area Perovskite Solar Module Fabrication. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74469>
11. Qi, J. et al. Geometrically Asymmetric Phosphonium Zwitterions Enable Efficient Printable Mesoscopic Perovskite Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74629>
12. Vaitukaitytė, D. et al. Anchor Engineering of Hole-Selective Enamine-Based Self-Assembling Monolayers for Perovskite Solar Cells. *Advanced Science* (2026). <https://doi.org/10.1002/advs.76950>
13. Xu, Y. et al. Closed-Loop Slit-Flushing Antisolvent Strategy Enables Controlled Three-Dimensional Mass Transfer for Large-Area Perovskite Solar Cells. *Small* (2026). <https://doi.org/10.1002/sml.75068>

14. Kang, R. et al. Thiourea Derivatives for Mitigating Photodegradation-Mediated Lead Leakage in Sustainable Perovskite Solar Cells and Modules. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acsenergylett.6c02033>
15. Wang, Z. et al. Synergistic Self-Passivation of Vacancy Complexes in Metal Halide Perovskites. *ACS Nano* (2026). <https://doi.org/10.1021/acsnano.6c10992>
16. Ouyang, D. et al. Diminishing Defect and Ion Migration in Perovskite-Organic Tandem Solar Cells by Ordered Anion-Cation Modification Approach. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77855>
17. Gao, Z. et al. Selenocyanate Driven Organic Cations Homogenization in Wide-Bandgap Perovskite. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71462>
18. Chu, H. et al. Polyfluorinated-Regulator-Assisted Scaffold-Directed Tilted 1D/3D Heterointerfaces for High-Voltage and Stable Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.8256065>
19. Rout, P. et al. Halide Exchange via Perovskite Nanocrystals for Interfacial Defect Passivation in Solar Cells. *ChemRxiv* (2026). <https://doi.org/10.26434/chemrxiv.15007814/v1>
20. Huang, J. et al. Sustainable Stabilization of Perovskite Solar Cells Using Multifunctional Ammonium Alginate for over 26% Efficiency. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.78067>
21. Zhang, X. et al. Multiple Passivation on High-Quality Perovskite Crystals Based on Low-Cost Polyfunctional Food Additive for High-Performance Perovskite Solar Cells in Ambient Air. *ACS Applied Materials & Interfaces* (2026). <https://doi.org/10.1021/acsami.6c10456>
22. Wang, C.-. et al. Sulfamethoxazole-Induced Synergistic Interfacial Passivation for Efficient and Stable Perovskite Solar Cells. *The Journal of Physical Chemistry Letters* (2026). <https://doi.org/10.1021/acs.jpcllett.6c02253>
23. Duan, D. et al. Ionic Liquid Engineering Induced Hydrogen-Bond Networks for Efficient Wide-Bandgap Perovskite Photovoltaics. *Solar RRL* (2026). <https://doi.org/10.1002/solr.70461>
24. Chen, L. et al. Dual-Action Sulfonium-Amino Acid Additive Enabling Crystallization Regulation and Defect Passivation for High-Performance Carbon-Electrode Printable Mesoscopic Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.5667084>
25. Liu, X. et al. Novel Multi-Pyridyl-Functionalized Spirobifluorenes as Precursor Additives for Improving Efficient and Stable Perovskite Solar Cells. *ACS Applied Materials & Interfaces* (2026). <https://doi.org/10.1021/acsami.6c13903>
26. Lin, S. et al. Steric Hindrance-Assisted Coordination for Efficient and Stable Tin-Based Perovskite Solar Cells. *Solar RRL* (2026). <https://doi.org/10.1002/solr.70454>
27. Liang, J. et al. Solid-solution-induced lattice hardening reinforces perovskite stiffness for stable solar cell. *National Science Review* (2026). <https://doi.org/10.1093/nsr/nwag484>
28. Zhao, J. et al. Dynamic Strain Homogenization Regulation for Thermomechanically Robust Sn Pb Perovskite Solar Cells. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71497>
29. Sun, W. et al. Competitive Ligand Modulation Reduces Buried Interfacial Voids and Defects Enabling 26.17% Efficient Perovskite Solar Cells. *Small* (2026). <https://doi.org/10.1002/sml.75360>
30. Zha, L. et al. A Buried Sulfonate Molecular Bridge for Synchronous Charge Transport and Defect Passivation in High-Performance Perovskite Solar Cells. *Nano-Micro Letters* (2026).

- <https://doi.org/10.1007/s40820-026-02278-6>
31. Guo, H. et al. Molecular Disaggregation-Bridging Engineered SnO₂ Interfaces for Scalable Perovskite Photovoltaics. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71419>
 32. Hu, C. et al. Hydrogen-bond network-reinforced self-assembled monolayer enables high efficiency and operational stability in pure-iodide wide-bandgap perovskite solar cells. *DeCarbon* (2026). <https://doi.org/10.1016/j.decarb.2026.100190>
 33. Zhang, W. et al. Dipole-engineered co-adsorbed self-assembled monolayers boost Cs-based inorganic Pb-Sn perovskite tandem efficiency beyond 20%. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-76847-z>
 34. Zeng, X. et al. Dipole-Moment-Tuned Self-Assembled Monolayers for Efficient Perovskite Solar Cells. *ACS Applied Materials & Interfaces* (2026). <https://doi.org/10.1021/acsami.6c11467>
 35. Wang, H. et al. Dual Modulation of the Buried Interface in Inverted Perovskite Solar Cells With Heteroatom-Conjugated Asymmetric Self-Assembled Molecules. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77712>
 36. Qin, X. et al. Molecular Regulation toward SAM/Perovskite Interface for Efficient p-i-n Perovskite Solar Cells. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71482>
 37. Yang, J. et al. Efficient and Stable Inverted Perovskite Solar Cells Via a Multi-Arm Donor–Acceptor Dipole Molecular Bridge. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.3716107>
 38. Yuan, S. et al. Molecularly Templated Buried Interfaces for Inverted Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.6932848>
 39. Lu, C. et al. Near-Surface Sulfidation-Induced p-n Junction Enabling High-Efficiency Inverted Perovskite Solar Cells. *ACS Applied Materials & Interfaces* (2026). <https://doi.org/10.1021/acsami.6c07884>
 40. Zhu, Y. et al. Bifunctional NiO_x/Perovskite Interface Engineering for Efficient Air-Processed Wide-Bandgap Perovskite Solar Cells. *ACS Applied Energy Materials* (2026). <https://doi.org/10.1021/acsaeam.6c01708>
 41. Li, Y. et al. Structural Compensation Strategy: Composite NiO_x Hole Transport Layers Enabling High-Efficiency and Scalable Inverted Perovskite Solar Cells. *Small* (2026). <https://doi.org/10.1002/sml.75056>
 42. Ding, S. et al. Inhibiting Buried Mechanical Failure via Vapor-Induced Chemical Reconstruction for Durable Solar Cells. *Small* (2026). <https://doi.org/10.1002/sml.75122>
 43. Wang, X. et al. Clean Tin Oxide via Volatile Additive Engineering for High-performance Perovskite Photovoltaics. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71514>
 44. Zhou, S. et al. The Impact of C60-Self-Assembled Monolayer Electron-Transport Layers in Negative–Intrinsic–Positive Perovskite Solar Cells. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acsenergylett.6c01485>
 45. Bi, L. et al. Crystallization Modulation for Stable Wide-Bandgap Perovskite Solar Cells and Modules. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74388>
 46. Ali, W. et al. Suppressing Non-Radiative Recombination in Perovskite Solar Cells Through Integrated Tuning of Out-of-Plane Lattice Texture and Strain. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71474>
 47. Jiang, J. et al. Macroporous PbI₂ template and dual-mode coordination enable efficient perovskite solar cells. *Journal of Colloid and Interface Science* (2026). <https://doi.org/10.1016/j.j>

- jcis.2026.141449
48. Liu, T. et al. Fluorinated Phenethylammonium and Pre-Annealing Static Stage Enable Intermediate-Mediated Stepwise Crystallization in Tin Perovskite Solar Cells. *ACS Applied Materials & Interfaces* (2026). <https://doi.org/10.1021/acsaami.6c11057>
 49. Kitamura, T. et al. Solvent-Dependent Role of SnF₂ in DMSO-Free Tin Halide Perovskite Solar Cells. *ACS Applied Energy Materials* (2026). <https://doi.org/10.1021/acsaem.6c02063>
 50. Yin, Z. et al. Antiphase Boundaries Regulate Phase Stability and Performance in DMA + -Assisted CsPbI₃-Based Perovskites. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74538>
 51. Mikuciunaite, J. et al. Fluorinated Methylammonium Cation Containing Perovskite Solar Cells With Over 25% Power Conversion Efficiency. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71396>
 52. Yang, Y. et al. Precise Management of n-Value Distribution at the 2D/3D Interface Enabling Efficient Wide-Bandgap Perovskite Solar Cells. *Advanced Science* (2026). <https://doi.org/10.1002/advs.76791>
 53. Thor, W. et al. Diketopyrrolopyrrole-Based Cation as a Semiconducting Spacer in Layered Perovskite. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74681>
 54. Jouane, Y. et al. Stable lead-free bismuth perovskite solar cells via thermally activated bilayer conversion. *Solar Energy* (2026). <https://doi.org/10.1016/j.solener.2026.114961>
 55. Glazer, K. et al. Site-selective alkali ion alloying in double perovskites via liquid-assisted mechanosynthesis revealed by solid-state NMR. *ChemRxiv* (2026). <https://doi.org/10.26434/chemrxiv.15008057/v1>
 56. Padova, P.D. et al. Eco-friendly water-synthesis of lead-free Cs₂AgBiBr₆ perovskites. *Communications Materials* (2026). <https://doi.org/10.1038/s43246-026-01261-3>
 57. Gong, H. et al. Rationally Designed Cooperative Self-Assembled Monolayers Enable Holistic Modulations of Wide-Bandgap Perovskites for High-Performance Perovskite-Silicon Tandem Solar Cells. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71512>
 58. Park, S.J. et al. Self-Assembled Monolayers for High-Efficiency Perovskite/Si Tandem Solar Cells. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acsenergylett.6c01966>
 59. Mahmoud, M.A.A. et al. Impact of silicon substrate topography on sequentially evaporated perovskite film growth. *Joule* (2026). <https://doi.org/10.1016/j.joule.2026.102646>
 60. Zhao, Z. et al. High-Performance Perovskite/Silicon Tandem Solar Cells via Ion Modulation. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77637>
 61. Zhang, T. et al. Dimensional Seed Engineering of Strain in Wide-Bandgap Perovskites for Efficient and Stable Tandem Solar Cells. *ACS Nano* (2026). <https://doi.org/10.1021/acsnano.6c07355>
 62. Yuan, J. et al. Universal Sacrificial Coordination Strategy for ALD-Resilient SAMs Achieving High-Performance Perovskite/Organic Tandem Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74461>
 63. Bristow, H. et al. Material Selection through Accelerated Aging Enables >95% Performance Retention of a Perovskite–Silicon Tandem Solar Cell after 1 Year of Outdoor Monitoring. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acsenergylett.6c01363>
 64. Li, L. et al. Large-Scale Perovskite Photovoltaics: Mitigating Defect Challenges in Upscaled Modules and Monolithic Tandem Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.202532038>
 65. Zhang, J. et al. Decoupling Bidirectional Photochemical Degradation via Radical Scavenging

- for Stable Inverted Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.4221605>
66. Yan, X. et al. Self-assembled monolayers with cross-linkable header on the linker for durable perovskite photovoltaics. *Science Advances* (2026). <https://doi.org/10.1126/sciadv.aec3096>
 67. Luo, C. et al. Symmetry-breaking co-assembly of conjugated molecules boosts perovskite photovoltaics. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-76848-y>
 68. Guo, Y. et al. Regulating Molecular Adsorption Configuration via Spatially Balanced Polarity for High-Performance Inverted Perovskite Solar Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77942>
 69. Wang, K. et al. Acid-Base Complexation Induced Dipole Engineering for Durable Inverted Perovskite Photovoltaics. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74438>
 70. Tang, Z. et al. Mitigating Stress-Induced Nonphotoactive Phase Transition Through Sodium Sulfonate Engineering for Stable and Efficient Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.2425823>
 71. Ma, X. et al. Regulating lattice evolution via an ordered two-dimensional intermediate phase for resilient perovskite solar cells. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-76425-3>
 72. Abbas, Z. et al. Suppression of Lattice Heterogeneity for Efficient Inverted Perovskite Solar Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77729>
 73. Lukas, T. et al. Beyond the Gold Standard: Towards Industrially Viable Electrodes for Durable Perovskite Solar Cells. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71372>
 74. Liu, N. et al. Two Birds With One Stone: Modifier Interfacial Engineering for High-Efficiency Stable p-i-n All-Inorganic Perovskite/Organic Tandem Solar Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77805>
 75. Ren, H. et al. Pushing the Interconnection Limit in Perovskite-Organic Tandem Solar Cells Through Recombination Channels Delocalization. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.6548693>